

Gengcong Yang

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EDUCATION

University of Texas at Austin (UT Austin) <i>Master in Computer Science</i>	08/2023 – 12/2024 Austin, United States
Tsinghua University (THU) <i>Master in Control Engineering, Department of Automation Supervisor: Yujiu Yang</i>	08/2019 – 06/2022 Beijing, China
National Chiao Tung University (NCTU) <i>Exchange Student in Computer Science</i>	09/2017 – 01/2018 Hsinchu, Taiwan
Sun Yat-sen University (SYSU) <i>Bachelor in Software Engineering</i>	08/2015 – 06/2019 Guangzhou, China

PUBLICATIONS

Diamond: Harnessing GPU Resources for Scientific Deep Learning	2025
- Authors: Haotian Xie, Rohan Marwaha, Minu Mathew, Song Bian, Gengcong Yang, et al. - Contributed to the design of a distributed job-submission pipeline enabling scalable, multi-node GPU training on a platform for the scientific machine learning lifecycle; funded by NSF and published in eScience.	
Probabilistic Modeling of Semantic Ambiguity for Scene Graph Generation	2021
- Authors: Gengcong Yang, Jingyi Zhang, Yong Zhang, Baoyuan Wu, Yujiu Yang - Proposed a probabilistic framework and designed a plug-and-play uncertainty modeling module to capture semantic ambiguity and multiple plausible visual relationships in scene graphs. - Published in CVPR 2021.	

RESEARCH EXPERIENCE

Texas Advanced Computing Center, Graduate Research Assistant	Austin, 01/2024 – 12/2024
- Supervisor: Yinzhi Wang - Contributed to the early-stage development of Diamond, an NSF-funded, scalable platform supporting the full machine learning research lifecycle on HPC resources. - Designed and implemented a distributed job-submission pipeline enabling users to containerize environments and train ML models across multiple GPU nodes.	
KAUST, Research Intern	Remote, 02/2021 – 07/2021
- Supervisor: Bernard Ghanem - Implemented a differentiable RoI Pooling method with CUDA and C++ for the temporal action detection task on videos, achieving an experimental improvement of 6%.	
Tencent AI Lab, Computer Vision Researcher Intern	Shenzhen, 07/2019 – 07/2020
- Supervisor: Baoyuan Wu - Developed a novel modeling method for polysemous visual-relationship representation and applied for a technology patent. - Contributed to the design and experimentation of the probabilistic uncertainty modeling framework later published at CVPR 2021.	

INDUSTRY EXPERIENCE

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- TikTok USDS**, Emerging Product, *Machine Learning Engineer* San Jose, 07/2026 – Present
- Coordinated model evaluation for Seedance, a video generation model, ahead of U.S. rollout.
 - Supported model serving infrastructure for U.S. deployment, partnering with internal and external cloud providers on provisioning and monitoring.
 - Developed agents to automate the team’s operational workflows.
- Uber**, Driver Offer Pricing Team, *Machine Learning Engineer* Sunnyvale, 01/2025 – 07/2026
- Developed driver fare prediction models used as core pricing signals within Uber’s marketplace pricing system.
 - Improved observability and reliability infrastructure for pricing prediction pipelines running at scale.
- ByteDance**, E-commerce, *Machine Learning Engineer* Shanghai, 06/2022 – 06/2023
- Developed image search models and a retrieval-ranking framework for the TikTok Shop similar-product search project.
 - Constructed large-scale datasets for post-pretraining and quantized object detection models to improve serving efficiency.
- Tencent**, *Machine Learning Engineering Intern* Beijing, 06/2021 – 08/2021
- Focused on the retrieval stage of the recommendation system for an early-stage short-video platform.
 - Built an offline evaluation pipeline for retrieval models and experimented with pairwise training for a two-tower model.

PATENTS

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- Visual Relationship Recognition Method Based on Artificial Intelligence**
- Inventors: **Gengcong Yang**, Yong Zhang, Baoyuan Wu, Yanbo Fan, Zhifeng Li, Yujiu Yang, Wei Liu
 - Patent Number: CN202011108484
 - Granted Date: 03/05/2024

SELECTED AWARDS

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| Tsinghua SIGS Professional Practice Scholarship | 2022 |
| Tsinghua Excellent Scholarship First Prize | 2021 |
| SYSU Outstanding Undergraduate Thesis Award | 2019 |
| SYSU Excellent Scholarship Second Prize | 2018 |
| National Encouragement Scholarship | 2017, 2018 |
| SYSU Xian Weijian Overseas Study Funding | 2017 |
| Interdisciplinary Contest In Modeling Honorable Mention | 2017 |
| SYSU Excellent Scholarship Third Prize | 2016, 2017 |